



A Novel Test for MOND: Gravitational Lensing by Disk Galaxies

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Abstract

Disk galaxies represent a promising laboratory for the study of gravitational physics, including alternatives to dark matter (DM), owing to the possibility of coupling rotation curves' dynamical data with strong gravitational lensing (SGL) observations. In particular, Euclid, DES, and LSST are predicted to observe hundreds of thousands of gravitational lenses. Here, we investigate disk galaxy SGL in the modified Newtonian dynamics (MOND) framework. We employ the concept of equivalent Newtonian systems within the quasi-linear MOND formulation to make use of the standard lensing formalism. We derive the phantom DM distribution predicted for realistic disk galaxy models and study the impact of morphological and mass parameters on the expected lensing. We find purely MONDian effects dominate the lensing—due to the presence of a phantom DM disk—and generate nontrivial correlations between the lens parameters and the lensing cross section. Moreover, we show that MOND predicts a substantial increase in the number count of disk galaxy lenses compared to the DM-driven predictions, making it distinguishable from the latter in upcoming surveys regardless of the chosen interpolating function. Finally, we argue that disk galaxy gravitational lensing, coupled to additional astronomical observations, can be used to constrain the interpolating function of MOND.

Unified Astronomy Thesaurus concepts: Modified Newtonian dynamics (1069); Strong gravitational lensing (1643); Gravitation (661); Disk galaxies (391)

1 Introduction

The dark matter (DM) hypothesis is an essential element of the standard cosmological model, Λ CDM; see, e.g., Planck Collaboration et al. (2020). DM was originally introduced to explain the observed rotation curves of disk galaxies (A. Bosma 1978; V. C. Rubin et al. 1978; Y. Sofue & V. Rubin 2001): both the lack of the expected quasi-Keplerian falloff in the rotational velocity curve at large distances and the rotational velocity amplitudes pointed to an additional but invisible source of gravity, i.e., DM.

DM can successfully explain a plethora of astrophysical and cosmological observations, such as the cosmic microwave background power spectrum (G. Hinshaw et al. 2009) and gravitational lensing by galaxies and galaxy clusters (see, e.g., M. Bartelmann 2010; T. Treu 2010). In particular, DM halos are believed to dominate strong gravitational lensing (SGL) by galaxies (P. Schneider et al. 2006).

However, the results of the many experiments aimed at the direct detection of DM particles are, to date, inconclusive (see, e.g., J. Liu et al. 2017; Y. Meng et al. 2021; C. Blanco & R. K. Leane 2024). Furthermore, in the past two decades, the very nature of DM has become a point of strong debate within the scientific community (G. Bertone & D. Hooper 2018; S. Profumo et al. 2019).

Various alternative explanations of astrophysical observations have been proposed in the form of novel theories of gravity: modified gravity (J. W. Moffat 2006; J. W. Moffat & S. Rahvar 2013), fractional gravity (A. Giusti 2020; F. Benetti et al. 2023), rainbow gravity (J. Magueijo & L. Smolin 2004), emergent gravity (E. Verlinde 2011; E. P. Verlinde 2017),

postquantum classical gravity (J. Oppenheim & Z. Weller-Davies 2023; J. Oppenheim et al. 2024), $f(R)$ gravities (T. P. Sotiriou & V. Faraoni 2010), among others, as well as general relativistic models of disk galaxies (D. Astesiano et al. 2022; W. Beordo et al. 2024; M. Galoppo & D. L. Wiltshire 2024; M. Galoppo et al. 2025).

Any viable alternative to DM must explain the phenomena naturally understood within the standard DM paradigm. Furthermore, observational procedures must be designed to differentiate between the many theories now present in the literature and rule out those that do not match the observations. In this regard, disk galaxies represent an ideal laboratory to design tests to investigate DM and differentiate it from its alternatives. Indeed, we recall that in recent years the coupling of dynamics and SGL data in massive disk galaxies has already been used, on the few systems observed, to accurately probe the nature of DM (C. M. Trott et al. 2010; A. A. Dutton et al. 2011; S. H. Suyu et al. 2012).

In this work, we investigate inclination effects in disk galaxy SGL in the modified Newtonian dynamics (MOND; M. Milgrom 1983) scenario (see, e.g., B. Famaey & S. S. McGaugh 2012, and I. Banik & H. Zhao 2022 for reviews). To date, MOND represents the most widely studied alternative to the DM paradigm as it has been shown to successfully explain a plethora of astrophysical observations, such as kinematics of galaxies (B. Famaey & S. S. McGaugh 2012; Y. Zhu et al. 2023), dynamics of wide binary stars (X. Hernandez et al. 2023; K.-H. Chae 2024), radial acceleration relation of galaxies (RAR, F. Lelli et al. 2017; T. Misteale et al. 2024a), orbital velocity of interacting galaxy pairs (R. Scarpa et al. 2022), early-Universe galaxy and cluster formation (S. S. McGaugh et al. 2024), and galaxy-scale gravitational lensing (M. Milgrom 2013; T. Misteale et al. 2024b).



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In the Lagrangian formulation of MOND by J. Bekenstein & M. Milgrom (1984), referred to as AQUAL, the linear Poisson equation linking the Newtonian gravitational potential, Φ_N , to the matter density, ρ , i.e.,

$$\Delta\Phi_N = 4\pi\rho \quad (1)$$

is replaced by the nonlinear partial differential equation (PDE)

$$\nabla \cdot (x) \nabla \Phi_M = 4\pi\rho \quad (2)$$

where $x = \nabla \Phi_M$, $a_0 \approx 1.2 \times 10^{-13} \text{ km s}^{-2}$ is the MOND universal acceleration scale (B. Famaey & S. S. McGaugh 2012), Φ_M is the Milgromian gravitational potential and $\mu(x)$ is the MOND interpolating function, with the asymptotic limit behaviors

$$\mu(x) \approx \begin{cases} 1 & \gg 1 \\ x & \ll 1. \end{cases} \quad (3)$$

Equation (3) then implies that for $\nabla \Phi_M \gg a_0$, Equation (2) reduces to (1), and MOND recovers the standard Newtonian regime, while for $\nabla \Phi_M \ll a_0$ one obtains the deep MOND (dMOND) regime, practically defined by

$$\nabla \cdot \|\nabla \Phi_M\| \nabla \Phi_M = 4\pi\rho a_0 \quad (4)$$

The extent to which MOND effects dominate an astrophysical system can be roughly gauged by evaluating κ (B. Famaey & S. S. McGaugh 2012; I. Banik & H. Zhao 2022)

$$\kappa = \frac{M}{r_c^2 a_0} \quad (5)$$

where M is the system mass and r_c is the typical scale length of the system. When $\kappa \gg 1$, the system is in a Newtonian regime, while for $\kappa \ll 1$, dMOND applies. In particular, we recall that for disk galaxies $\kappa \approx 1$, so that these require the full use of Equation (2) to be described within the MOND scenario.

We note that for every astrophysical system modeled through MOND, it is possible to construct the equivalent Newtonian system (ENS), i.e., a system with the same baryonic matter density, ρ_B , supplemented by an additional halo of phantom dark matter (PDM), with density profile ρ_{ph} , such that $\Phi_N = \Phi_M$ throughout the physical system. In the AQUAL formulation of MOND, the construction of an ENS is a complex task due to the highly nonlinear nature of Equation (2).

However, M. Milgrom (2010) constructed a different formulation of MOND which allows for a more direct approach to building ENS, i.e., quasi-linear MOND (hereafter QUMOND). QUMOND essentially relies on an approximation of Equation (2) in which Φ_M is replaced by Φ_N . By this approach, QUMOND involves only the solution of linear PDEs, with only nonlinear algebraic steps. Indeed, in QUMOND, the Milgromian gravitational potential now solves for

$$\Delta\Phi_M = 4\pi[\rho_B + \rho_{ph}] \quad (6)$$

where ρ_{ph} is given by

$$\rho_{ph} = \frac{1}{4\pi} \nabla \cdot [\tilde{\nu}(y) \nabla \Phi_N] \quad (7)$$

where $y = \nabla \Phi_N$, and $\tilde{\nu}(y)$ has as asymptotic limiting behavior

$$\tilde{\nu}(y) \approx \begin{cases} 0 & \gg 1 \\ -1/2 & \ll 1. \end{cases} \quad (8)$$

Therefore, within the QUMOND approach, given ρ_B , the ENS is straightforwardly built by i) first solving the classical Poisson equation for the baryonic matter, which becomes the source of ρ_{ph} via a nonlinear algebraic procedure involving ν ; ii) then by solving once more the Poisson equation for $\rho = \rho_B + \rho_{ph}$.

Here, we stress that the PDM is not constituted by real particles, and thus does not lead to dynamical dissipation or friction (B. Famaey & S. S. McGaugh 2012; M. Bílek et al. 2021; P. Kroupa et al. 2023). Furthermore, we note that the PDM density can become negative locally and, although it will respect the symmetries of the baryonic distribution, it can present shifts in its concentration peaks with respect to the former. In particular, for flattened matter distributions such as disk galaxies, the local positivity of ρ_{ph} is not assured, nor is the correspondence between the peak of baryonic density and of PDM (M. Milgrom 1986; L. Ciotti et al. 2006; X. Wu et al. 2008; L. Ciotti et al. 2012; C.-M. Ko 2016). Through the use of the ENS, one can study lensing of disk galaxies in MOND, while utilizing the tools constructed for classical SGL within the Λ CDM paradigm. This allows us to bypass a series of difficulties in the study of SGL within MOND, e.g., the definition of the thin lens approximation, commonly employed for galaxy-scale SGL (D. J. Mortlock & E. L. Turner 2001). Furthermore, the ENS allows us to probe the Newtonian and “transition” regime of MOND in SGL, as well as directly gauge the relevance of the dMOND limit in lensing for highly nonspherical systems.

Here, we note that SGL has often been regarded as a secondary probe within the MOND paradigm, following order of magnitude estimates derived from spherically symmetric lenses in which PDM contributions to lensing are minor with respect to the baryonic contribution (see, e.g., M. Milgrom 2020). However, it has actually already been shown that even for spherically symmetric lenses, the PDM can contribute $\approx 15\%$ of the projected matter content within the associated Einstein radius, thus producing a nonnegligible effect on the SGL (R. H. Sanders 2014). In disk galaxy lenses, due to the presence of a PDM disk (see Figure 2), we might expect even higher contributions of the PDM to the projected mass within the Einstein radius, specifically as the inclination angle increases toward an edge-on configuration.

In this paper, we investigate inclination effects on the observables of SGL for disk galaxies, e.g., SGL cross section, within the MOND scenario. We draw comparison to known results on disk galaxy lensing in Λ CDM, and we show that: i) the number count of disk galaxy lenses predicted by MOND exceeds the one expected in the DM scenario, and ii) in MOND disk galaxy lensing, the inclination effects are enhanced with respect to the DM halo paradigm. With the upcoming optical surveys, LSST, DES, and Euclid, these differences in the inclination effects and overall lens number counts will become statistically detectable in SGL of disk galaxies, as these surveys are predicted to observe hundreds of thousands of galaxy-scale lenses (T. E. Collett 2015;

J. A. Acevedo Barroso et al. 2025). Thus, we propose that, in the next decade, statistical studies on the inclination and number of disk galaxy lenses will lead to an important direct test to differentiate cold DM from MOND.

Here, we focus on disk galaxies as isolated systems, assuming no interfering matter along the line of sight. This is a typical practice in studies of SGL on galaxy scales, where matter contributing to the lens effect from outside the lens galaxy is modeled by fitting an external convergence parameter, often in combination with weak lensing measurements see, e.g., O. Tihhonova et al. 2018; P. Wells et al. 2023). However, we note that due to external field effects (EFE) in the MOND paradigm, the assumption of an isolated system is ultimately a stronger requirement than in the DM scenario B. Famaey & S. S. McGaugh 2012; I. Banik & H. Zhao 2022). Indeed, due to the nonlinear nature of the MOND field equation see Equations 2) and 6)), the presence of external matter nontrivially influences the internal gravitational field of the lens itself. Therefore, our study will be directly applicable to highly isolated disk galaxies, while case-by-case corrections for EFE will have to be considered for SGL within clusters and more nuanced environments.

Throughout this paper, we assume a spatially flat Λ CDM cosmology, with matter density $\Omega_{M0} = 0.3$ and Hubble constant $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, as a background cosmological model, with lens and source placed at redshift $z_L = 0.5$ and $z_S = 2.0$, respectively. We note that assuming a spatially flat Λ CDM cosmology is not at odds with a MOND description of disk galaxies. Indeed, even if MOND were to be ultimately correct, its relativistic extension would have to closely match the cosmological evolution predicted by the highly tested Λ CDM model. Therefore, employing Λ CDM as a cosmological model within a MOND scenario at the galactic scales can be directly understood as a good first-order approximation to the possible choice of a correct, and yet unknown, MOND cosmology B. Famaey & S. S. McGaugh 2012; I. Banik & H. Zhao 2022).

The rest of the paper is structured as follows. In Section 2 we recall the SGL fundamentals and derive the observables used to probe inclination effects in disk galaxy SGL. In Section 3 we define the analytic baryon density distribution for the disk galaxy models, and we derive the respective PDM distribution for realistic templates of such galaxies. In Section 4 we report our results, focusing on the effect of disk galaxy inclination on SGL cross sections and draw a comparison to the DM scenario. Finally, Section 5 is dedicated to conclusions and future perspectives.

2 Strong Gravitational Lensing

For a given density distribution, in the weak-field limit of gravitational fields, the deflection angle of a light ray propagating perpendicular to the plane of the lens, α , is defined as¹

$$\alpha(\xi_1, \xi_2) = \frac{2}{c} \int_{-\infty}^{\infty} d\xi_3 \, \perp(\xi). \quad (9)$$

¹ Employing 9) implies a restriction on the possible choice of the a priori unknown relativistic MOND extension. Indeed, we are assuming that the lensing potential is twice the Newtonian potential in the weak-field regime, as in general relativity see, e.g., J. D. Bekenstein 2004 for such a theory).

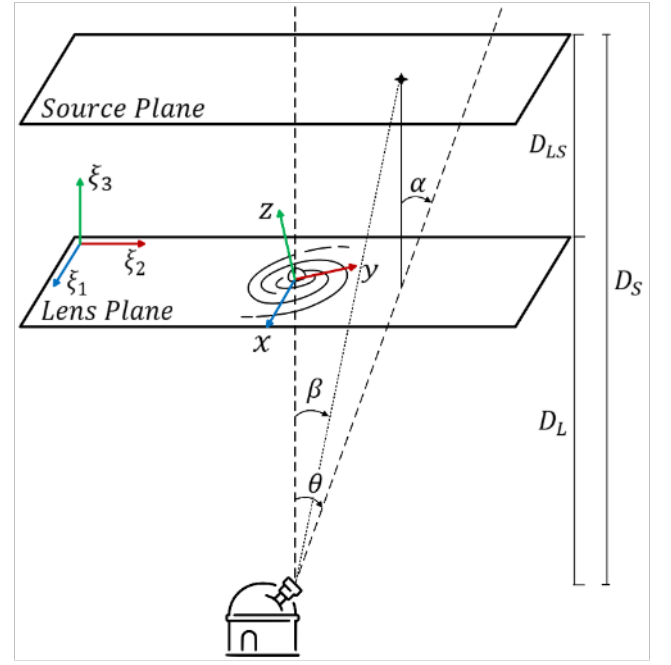


Figure 1 Lensing configuration for SGL by a disk galaxy. We report source, lens, and observer positions, as well as the relevant angles and coordinate systems. Here, α is the deflection angle, β is the angular source position, and θ is the angular image position.

Here, $\perp$ denotes the acceleration due to the gravitational potential of the lens object along the lens plane, i.e.,

$$\perp = -\nabla_{\perp} \Phi_N \quad (10)$$

with

$$\Phi_N(\xi) = \iiint \frac{\rho(\xi')}{|\xi - \xi'|} d^3\xi' \quad (11)$$

It is important to stress that the lens plane lies orthogonal to the line of sight from the observer to the lens and thus, it is not dependent on the inclination of the lens itself see Figure 1). Therefore, the inclination of a disk galaxy will have a large impact on its lensing, which is derived from the projection of its baryonic matter content onto the lens plane. Here, we note that in the presence of (almost) spherical DM halos, the inclination effects of disk lenses, although still clearly present, are partially suppressed by the role played by the DM in the lensing phenomena C. R. Keeton & C. S. Kochanek 1998). However, we do not expect such a situation in MOND, where no DM halos are present and the lensing is driven solely by the baryonic disk.

We note that by employing Equations 9)–11) coupled to the ENS of the disk galaxy's gravitational potential, SGL in MOND is then easily integrated into the usual lensing formalism, with Φ_M in place of the classical Newtonian potential Φ_N .

We can now follow the standard lensing formalism see, e.g., P. Schneider et al. 2006; M. Meneghetti 2022) to define the lens equation

$$\beta = \theta - \frac{LS}{s} \alpha(\theta) = \theta - \alpha(\theta) \quad (12)$$

where β and θ , respectively, describe the angular position with respect to the lens center of the source object and of the lensed images. Moreover, D_{LS} and D_S represent the angular diameter distances from the lens to the source and from the observer to the source. Figure 1 depicts the lensing configuration under study. Then, from (12) we can define the Jacobian matrix

$$\frac{\partial \beta}{\partial \theta} = -\nabla \alpha \quad (13)$$

The critical curves for the lens configuration are defined as the positions where the magnification, $\mu := \det J$, is formally infinite (for a point source). We note that it is, however, generally convenient to determine critical curves directly from the condition $\det J = 0$. Then, we can use the lens Equation (12) to project the critical curves back onto caustic curves on the source plane.

The position of a source object with respect to the caustic lines determines the number of images produced and their respective positions. Objects within caustic lines will produce multiple images, and thus, the area enclosed by caustic lines on the source plane defines the lensing cross section σ (P. Schneider et al. 2006; M. Meneghetti 2022). We note that while the lensing cross section is a survey-independent quantity, the likelihood of observing SGL is not. The latter, although directly proportional to the former, will be influenced by the specific observations carried out (T. E. Collett 2015; M. Meneghetti et al. 2022). Therefore, to characterize the inclination effects of disk galaxies in MOND, we only consider σ as an indirect measure of SGL likelihood and do not specialize in any specific surveys.

Finally, in this paper, we focus on point sources. Extended sources will have higher lensing likelihood, but for the purposes of determining the differences between lensing effects for different disk galaxy mass profiles and inclination effects, considering pointlike sources is sufficient.

3 Matter Density Profiles

To describe the baryonic component of disk galaxies, we couple a thick exponential disk to a spherical bulge, such that $\rho_B = \rho_b + \rho_d$, where ρ_b and ρ_d are the densities of the bulge and the disk, respectively.

3.1 Analytical Baryonic Matter Density Modeling

The spherical bulge component is well described by the Plummer density profile (E. Poulidas et al. 2017)

$$\rho_b(r) = \frac{3r_b^2}{4\pi r^2} \frac{b}{(r_b^2 + r^2)^{5/2}} \quad (14)$$

where r is the spherical radius, r_b is the scale parameter of the bulge, and M_b is the total bulge mass. The baryonic disk matter distribution is conventionally taken as $\rho_d(R, z) = \Sigma(R)Z(z)$, where R is the cylindrical radius and z is the axial coordinate. Then, the surface density of the disk is well approximated by the exponential profile (see, e.g., J. Binney & S. Tremaine 2008)

$$\Sigma(R) = \frac{d}{2\pi R_d^2} e^{-R/R_d} \quad (15)$$

where M_d is the total disk mass and R_d is the scale length in the radial direction for the galactic disk. The distribution along z ,

such that $\int Z(z) dz = 1$, can be defined as (J. Binney & S. Tremaine 2008)

$$Z(z) = \frac{1}{2z_d} e^{-|z|/z_d} \quad (16)$$

where z_d is the thickness scale length of the disk galaxy. In separating the density profile in this way, we have assumed that the disk has a scale height that is constant with radius, as is observed in the photometry of local edge-on disk galaxies (P. C. van der Kruit & L. Searle 1981, 1982). Thus, we have

$$\rho_d(R, z) = \frac{d}{2\pi R_d^2 z_d} e^{-R/R_d} e^{-|z|/z_d}. \quad (17)$$

We recall that in the pioneering work of C. R. Keeton & C. S. Kochanek (1998) on SGL by disk galaxies within the Λ CDM paradigm, the baryonic disk component was approximated either by Mestel (L. Mestel 1963) or Kuzmin disks (G. G. Kuzmin 1956) within a softened, oblate isothermal DM halo.² Moreover, similar chameleon profiles were employed to combine dynamical and SGL data for disk galaxies in the more recent studies by A. A. Dutton et al. (2011) and S. H. Suyu et al. (2012), as well as in the first consideration of nonspherical lenses in MOND (H. Y. Shan et al. 2008). These profiles have been preferred to fully exponential mass distributions for the disk component due to their analytical properties in SGL computation (C. R. Keeton & C. S. Kochanek 1998; H. Y. Shan et al. 2008; A. A. Dutton et al. 2011; S. H. Suyu et al. 2012).

In MOND, the lensing is entirely defined by the baryonic components of the galaxies, as no DM is required. Thus the morphological characteristics of the baryons in disk galaxies are more important in MOND than for cold DM scenarios. Furthermore, as already discussed by C. R. Keeton & C. S. Kochanek (1998), Mestel and Kuzmin disks represent a poor fit for the vertical component of the baryonic disk and can only be a first approximation for the exponential radial profile. Hence, to avoid biases in the inclination effects in MOND, we chose to maintain a closer match to observational data for the mass density profile modeling of disk galaxies and implement a realistic exponential disk profile coupled to a spherical bulge.

3.2 Realistic Disk Galaxy Model Parameters

Once the baryonic mass profile has been defined, we have to pick meaningful values for the parameters so that the resulting disk galaxy models can be understood to be a good representation of observational data.

Following C. R. Keeton & C. S. Kochanek (1998), we consider a massive spiral galaxy with a maximal disk, such that DM (or MOND) effects play a major role only in the outer kinematics of the galaxy. We therefore select a massive disk galaxy with $M_d = 10^{11} M_\odot$, and $R_d = 3.5$ kpc. We then consider three possible values for z_d , namely $z_d \in \{0.35, 0.105, 0.035\}$ kpc, so that the ratio

² We note that in the papers by Y. Wang & E. L. Turner (1997) and O. Möller & A. W. Blain (1998), uniform disks were considered for modeling the lens effect of a disk galaxy. However, due to the uniform assumption, and as noted by C. R. Keeton & C. S. Kochanek (1998), these cannot produce realistic rotation curves and do not depict actual three-dimensional density profiles of disk galaxies.

$z_d/R_d \in \{0.1, 0.03, 0.01\}$ as chosen by C. R. Keeton & C. S. Kochanek (1998), thus facilitating comparisons.

We must also select appropriate values for M_b and r_b . We note that correlations between M_b , r_b , and the disk parameters have been established in the literature—see, e.g., P. Lang et al. (2014; R. J. Allen et al. 2017; J. C. Forbes & A. Loeb 2018). However, these are better understood for low-mass spirals, and given the plethora of disk galaxy morphologies, these correlations produce only weak constraints on the bulge parameters once the disk has been selected. Therefore, in our analysis, we selected $M_b/M_d \in \{0.01, 0.1, 0.2\}$ and $r_b/R_d \in \{0.03, 0.1, 0.2\}$.

Finally, we point out that our choice of following C. R. Keeton & C. S. Kochanek (1998) in studying SGL by disk galaxies only in massive spirals is well justified by noting that the likelihood of SGL increases with the mass. Therefore, the low-mass spirals are not expected to significantly contribute to the population of disk galaxy lenses, even in the upcoming optical surveys.

3.3 Equivalent Newtonian System

To build the ENS corresponding to our choice of $\rho_B = \rho_d + \rho_b$ we exploit the linearity of the Poisson equation, so that

$$\Phi_N = \Phi_{N,d} + \Phi_{N,b} \quad (18)$$

where $\Phi_{N,d}$ and $\Phi_{N,b}$ are the Newtonian gravitational potentials generated by the exponential disk and the spherical bulge, respectively. We can directly calculate the two potentials by solving Equation (1) with (14) and (17) as sources. We then obtain

$$\Phi_{N,d}(R, z) = -M_d \int_0^\infty dk \frac{J_0(Rk) e^{-|z|k} - z_d k e^{-|z|k} z_d}{1 - R_d^2 k^2} \frac{1}{1 - z_d^2 k^2} \quad (19)$$

$$\Phi_{N,b}(r) = \frac{M_b}{r_b} \left(1 - \frac{r_b}{\sqrt{r_b^2 + r^2}} \right) \quad (20)$$

where J_0 is the Bessel function of the first kind. We can now directly calculate $\nabla\Phi$ and exploit Equation (7) to obtain ρ_{ph} once we have chosen a functional form for the interpolating function ν (see Appendix A). In this work, we mainly implement the same functional form used in the Phantom of Ramses code for N -body and hydrodynamical simulations in the MOND scenario³ (G. N. Candlish et al. 2014; F. Lüghausen et al. 2015), i.e.,

$$\tilde{\nu}(y) = -\frac{1}{2} \sqrt{\frac{1}{4} - \frac{1}{y^2}} \quad (21)$$

Here, it is important to note that while studies of disk galaxy rotation curves are quite robust with respect to the choice of interpolating function (due to the onset of the dMOND regime), in SGL, this choice is highly nontrivial. Indeed, SGL

effects probe the full spectrum of applicability of MOND and thus could be affected by the choice of ν .

In this paper, we have chosen as standard the widely adopted form of interpolating function of Equation (21) as it has been proven to give a MOND realization able to pass a variety of observational tests (B. Famaey & S. S. McGaugh 2012), and it is widely employed in the study of in-galaxy dynamics within the MOND scenario—see, e.g., F. Re & P. Di Cintio (2023; P. Di Cintio et al. 2024). However, the results may change with different choices of the interpolating function, as already noted in previous SGL MOND studies—see, e.g., R. H. Sanders & D. D. Land (2008). Therefore, we also implement a variety of alternative interpolating functions to: i) gauge the robustness of the results with respect to the choice of interpolating function; and ii) investigate the role of the transition acceleration regime probed by SGL. Specifically, we employ the RAR interpolating function—see, e.g., S. McGaugh et al. (2016; R. Stiskalek & H. Desmond 2023)—defined by

$$\tilde{\nu}_{RAR}(y) := \frac{-\sqrt{y}}{1 - \sqrt{y}} \quad (22)$$

as well as the sigmoid interpolating function employed by H. Desmond et al. (2024) to probe the role of the transition regime within solar system tests, i.e.,

$$\tilde{\nu}(y) := \Theta(\tilde{N}_{trans} - y) \tilde{\nu}_{RAR}(y) \quad (23)$$

where $\tilde{N}_{trans}$ is a fictitious transition scale such that $\tilde{N}_{trans} \geq 1$. Finally, we also implement an infinitely sharp transition interpolating function, namely

$$\tilde{\nu}(y) := \Theta(1 - y)^{-1/2} \quad (24)$$

to investigate directly the role of the dMOND regime within SGL of disk galaxies.

In Figure 2 we show as an example, the distribution of PDM—obtained employing the interpolating function of Equation (21)—from an edge-on and face-on point of view for a disk galaxy with $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. For the same galaxy and choice of interpolating function, in Figure 3 we show the ratio of the baryon density to the PDM density, ρ_B/ρ_{ph} , along the radial direction on the equatorial plane, and the corresponding two-dimensional distribution in the x - z plane within a scale height of the galactic disk. From Figure 2 we note that the resulting PDM distribution follows the same symmetries of the underlying baryonic matter, as we should expect from Equation (7). We find it to be always positive definite and with no peak shift in its concentration with respect to the baryons. Moreover, we recover the superimposed disk halo PDM morphology shown in previous studies—see, e.g., F. Lüghausen et al. (2015).

Furthermore, from Figure 3 we see that the PDM overtakes the baryonic matter as the dominant “matter,” component within the very core of the galaxy (below the parsec level) and at large distances from the galactic center (i.e., approximately at 15 kpc), precisely where we would expect MOND (or DM) effects to dominate the dynamics for this type of galaxy. The overtake of the PDM over the baryons within the center of the disk galaxy indicates the presence of a cusp in the PDM distribution. We note that such behavior is not related to the specific choice of interpolating function, as the center of the galaxy is in a dMOND regime, but relates to the interplay of

³ Here, we have implemented an original routine to derive the PDM distribution and the SGL effect of the disk galaxies. The codes implemented are available at https://github.com/chrisrhaw/MOND_lensing. Moreover, we give a description of these codes in Appendix B.

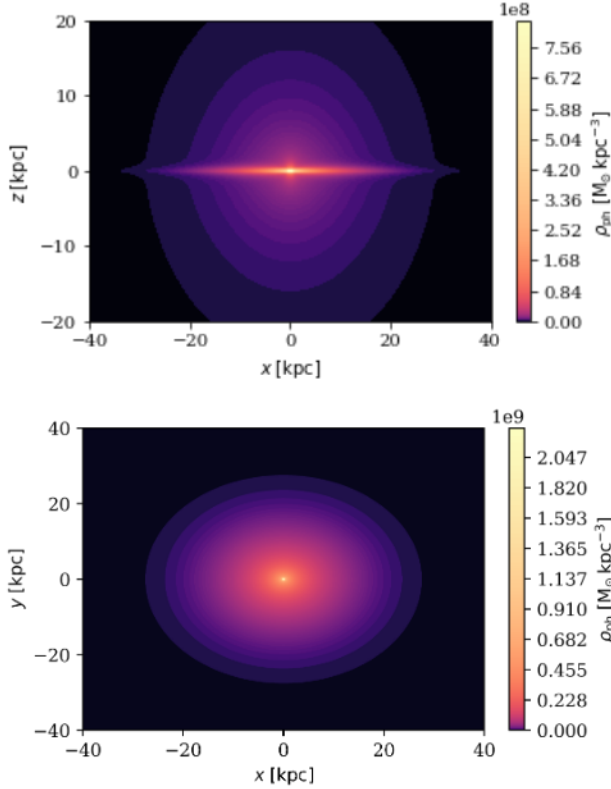


Figure 2 PDM density distribution, for the interpolating function of Equation (21), on the x - z plane (upper panel) and on the x - y plane (lower panel) for a disk galaxy with $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. A log scale is used for the density to highlight the morphology of the PDM distribution.

the latter with the baryon density profile. As such, it is a feature shared by any interpolating function that correctly encodes the dMOND limit.

Finally, we note that although the choice of interpolating function will impact the PDM distribution, we find that the PDMs are overall compatible with each other when probing the same acceleration regime. Indeed, although the PDM densities display differences, these display ratios of one with respect to the other only of ().

4 Results

In Figure 4 we show an example of the deflection angle, α , the magnification, μ , and the shear, γ , for the example of a massive disk galaxy with $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, $M_b = 10^9 M_\odot$, $r_b = 0.7$ kpc, and $i = 0^\circ, 90^\circ$. As expected, in the face-on case, the geometry of the lensing agrees with that of an axisymmetric lens, and moreover gives physically plausible deflection angles. This is consistent throughout the array of inclination angles, as can be seen for the edge-on case, in which the maximum value of α approximately doubles with respect to the face-on scenario. For $i = 90^\circ$, the breaking of the circular symmetry of the projected matter density on the lens plane drastically changes the geometry of the lensing effects, already showing the importance of inclination for disk galaxy lensing in MOND.

We show in Figure 5, for a fixed source position and lens parameters, that a wide variety of image configurations can be obtained by changing the inclination angle of the lens with

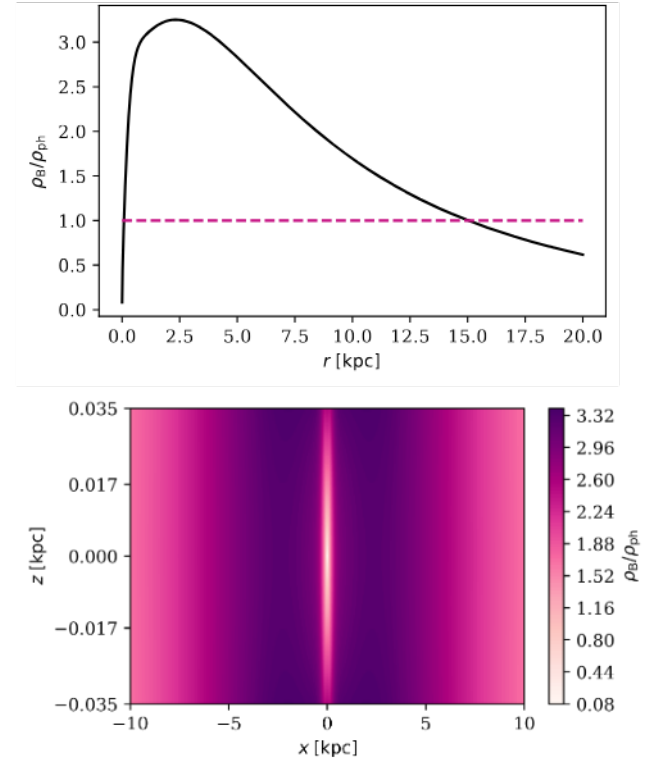


Figure 3 Ratio of the baryon density to the PDM density, ρ_B/ρ_{ph} , along the radial direction on the equatorial plane (top panel), and the corresponding two-dimensional distribution in the x - z plane within a scale height of the galactic disk (bottom panel) for a disk galaxy with $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The purple dashed line in the top panel highlights the point at which the two densities have the same value. The interpolating function employed here refers to that of Equation (21).

respect to the observer. Indeed, we recover familiar image configurations within our results—such as the Einstein cross quad-image configuration—already found in the previous studies on similar lensing systems. Here, we note that the tangential caustic curve of the lens does not develop butterfly features, even at high inclinations. This is in agreement with the results of C. R. Keeton & C. S. Kochanek (1998) for disk galaxies within the DM paradigm, and in contrast with the findings of Y. Wang & E. L. Turner (1997) and O. Möller & A. W. Blain (1998). However, the density modeling employed by these studies presents unphysical features which might then influence the morphology of the tangential curve (see G. Orban de Xivry & P. Marshall 2009 for a discussion on exotic caustic curve morphologies).

We find that changing the lens parameters, restricted to realistic disk galaxy models, does not significantly influence the shape of critical or caustic curves produced. However, varying the lens morphology and mass distribution in this way led to direct changes in the overall caustic area, as is shown in Figures 6 and 7.

In Figures 6–8 we show the lensing cross section with varying inclination angle and an ensemble of disk galaxy parameters. In these plots, due to the nature of calculating the SGL cross section via numerical methods on a grid, we assume a Poissonian error over the pixel count contained within the tangential caustic curves, i.e., the square root of the number of pixels multiplied by the single-pixel area value).

Figure 6 shows the effects of varying the disk thickness z_d on the lensing cross section. Interestingly, no clear trend with

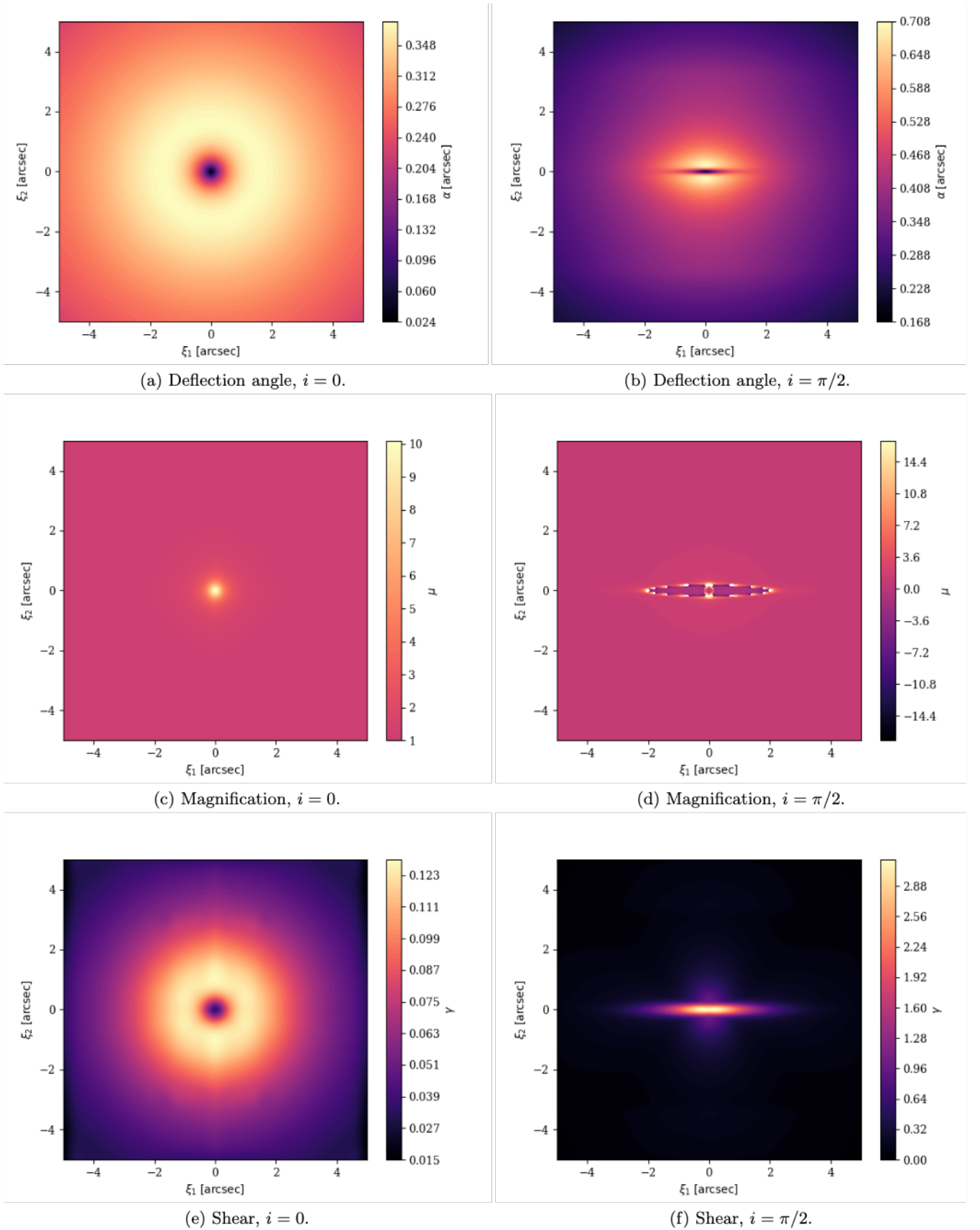


Figure 4 Example of lensing observables, namely the deflection angle (α), the magnification (μ), and the shear (γ), for a disk galaxy of inclinations $i = 0$ (left) and $i = \pi/2$ (right). The disk galaxy model is obtained by fixing $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc, and applying the interpolating function of Equation (21).

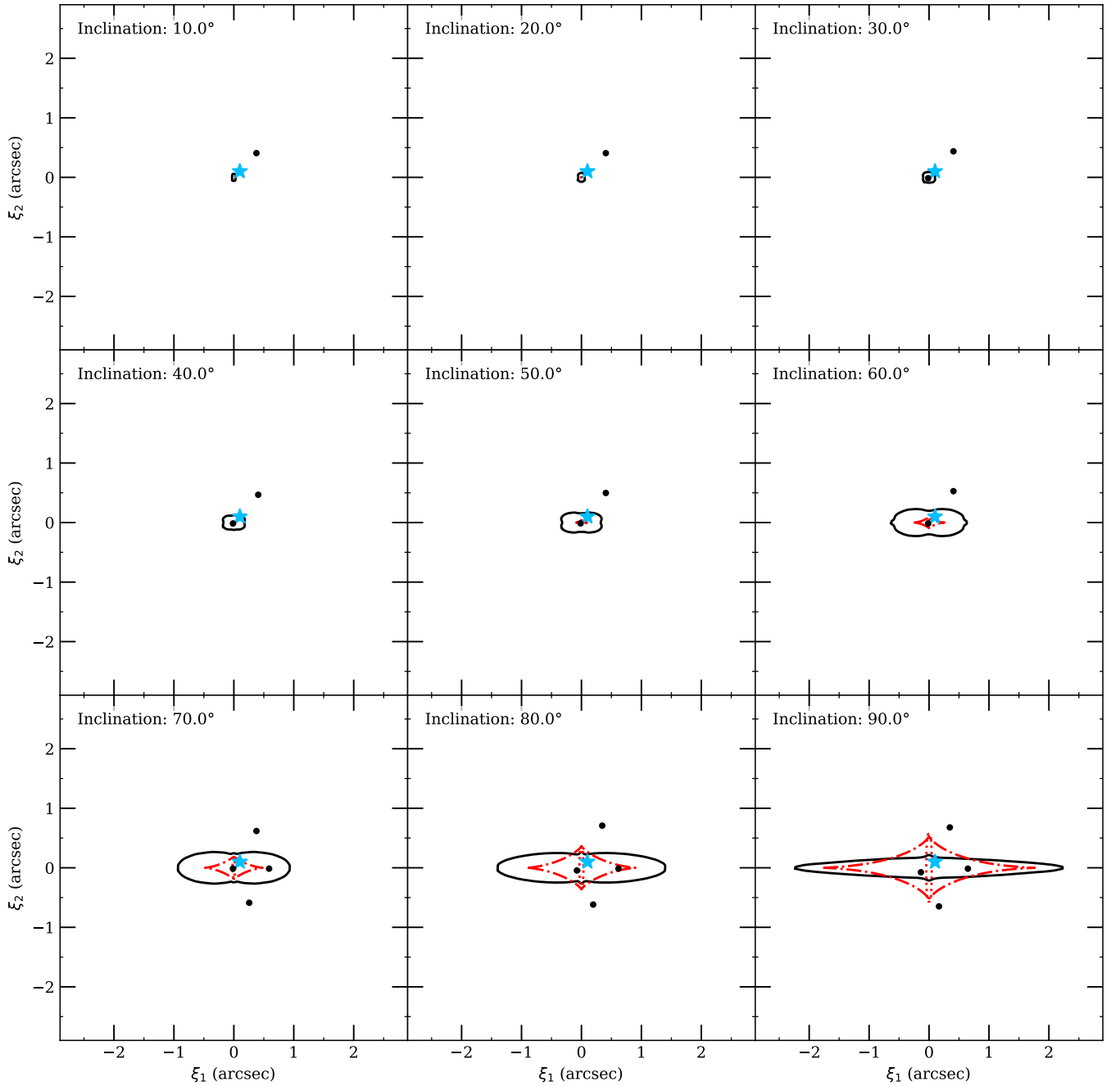


Figure 5 Example image configurations, caustic curves, and critical curves for spiral galaxy MOND lenses with varying inclination and point source objects placed at a constant position relative to the lens center. The solid black line, dotted red line, and dashed–dotted red line indicate the critical curve, radial caustic curve, and tangential caustic curves, respectively. The blue star represents the source image, and black dots show the corresponding lensed images. The parameters of the lens galaxy are fixed to be $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The interpolating function employed was the standard function of Equation 21). The lens and source are placed at redshift $z_L = 0.5$ and $z_S = 2.0$, respectively.

respect to the disk thickness emerges, already indicating the nontrivial interplay between nonlinear MONDian effects and lens parameters. Indeed, an increased lensing likelihood for more diffuse disk masses is in contrast to the standard lensing expectations. In the weak-field limit of general relativity, a more concentrated mass would produce stronger lensing effects.

In Figure 7 the nonlinear, non-Newtonian effects of MOND are more clearly captured. We see that an increased bulge mass negatively correlates with SGL cross section, within the mass range explored. Such a behavior is in stark contrast to

expectations from the standard weak-field SGL of a disk galaxy (C. R. Keeton & C. S. Kochanek 1998). This trend can be understood by noting that a more massive bulge, while enhancing the Newtonian lensing signal, can reduce the MONDian contribution by increasing the internal acceleration scale of the galaxy. We emphasize that this effect stems from the inherently nonlinear character of MOND and is therefore expected to be ultimately independent of the choice of interpolating function. Quantitatively, the analytical treatment of the modeled system (see Appendix A) shows that the bulge and disk can interact in a nontrivial way. For example, the

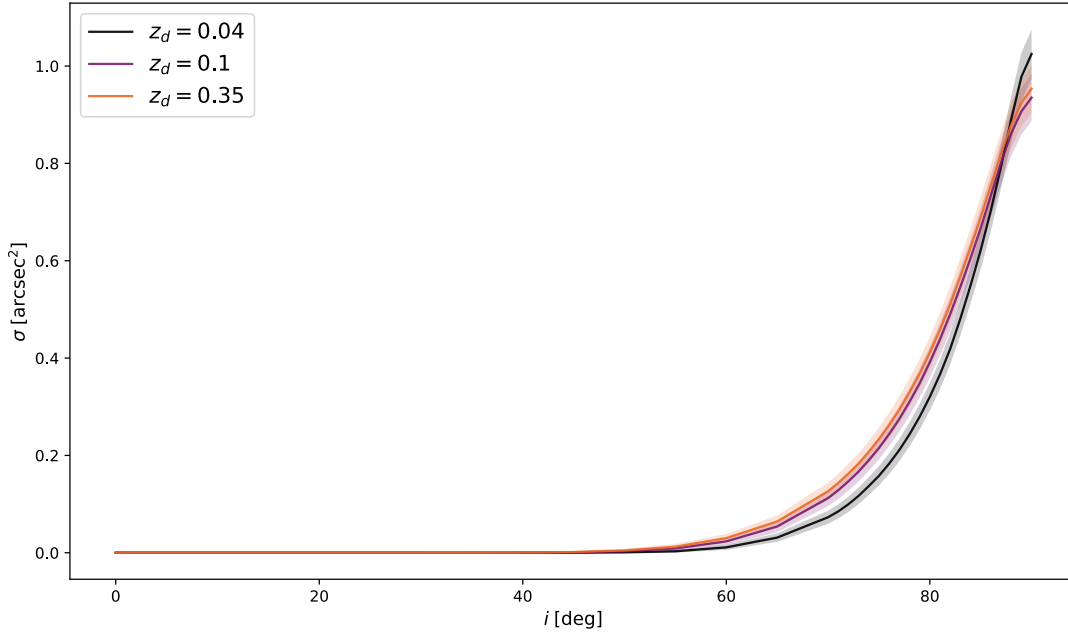


Figure 6 Impact of disk thickness z_d on lensing cross section σ with varying inclination angle within the MOND scenario. The other corresponding lens galaxy parameters are $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The lens and source are placed at redshift $z_L = 0.5$ and $z_S = 2.0$, respectively. The shaded areas represent the one standard deviation uncertainties around the obtained SGL cross section curves.

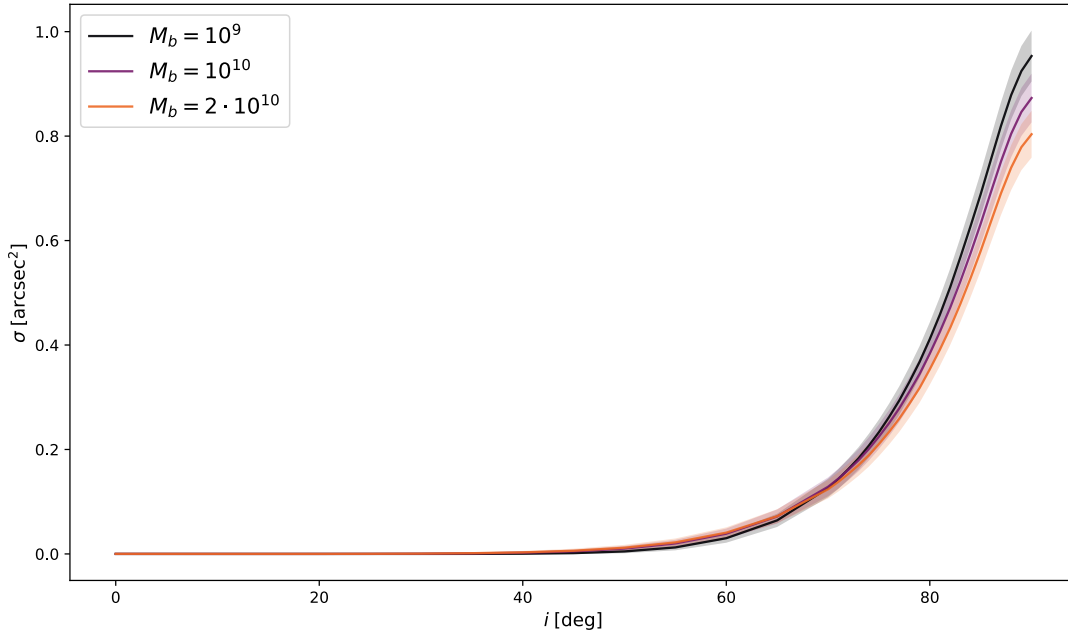


Figure 7 Impact of bulge mass M_b on lensing cross section σ with varying inclination angle within the MOND scenario. The other corresponding lens galaxy parameters are $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, and $r_b = 0.7$ kpc. The lens and source are placed at redshift $z_L = 0.5$ and $z_S = 2.0$, respectively. The shaded areas represent the one standard deviation uncertainties around the obtained SGL cross section curves.

disk-bulge interaction terms in Equations A7) and A8) can either enhance or suppress the PDM contribution to the total mass, once the morphological characteristics are specified.

Finally, in Figure 8 we present a comparison between the SGL cross section predictions for the same lens disk galaxy in the MOND scenario and the standard DM paradigm. The DM halo is assumed to be a conventional nonsingular isothermal

sphere NSIS; see, e.g., R. Kormann et al. 1994)

$$\rho_{\text{NSIS}}(r) = \frac{\rho_0}{2} \left(\frac{r_0}{r} \right)^2 \quad (25)$$

with parameters taken to be $\rho_0 = 2.51 \cdot 10^7 M_\odot \text{ kpc}^{-3}$, and $r_0 = 5.48$ kpc, for the template disk galaxy with

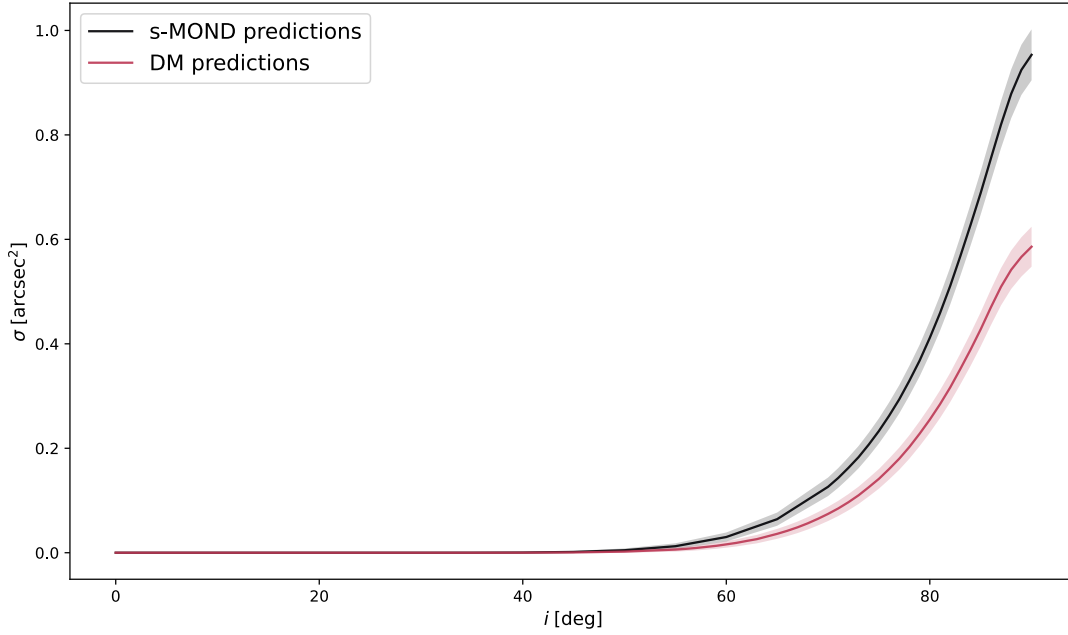


Figure 8 A comparison of lensing cross section predictions with varying inclination angle for the same disk galaxy in MOND and the conventional DM scenario. The disk galaxy parameters are $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, and $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The DM halo is taken as an NSIS profile with parameters $\rho_0 = 2.51 \cdot 10^7 M_\odot \text{ kpc}^{-3}$, and $r_0 = 5.48$ kpc. The lens and source are placed at redshift $z_L = 0.5$ and $z_S = 2.0$, respectively. The shaded areas represent the one standard deviation uncertainties around the obtained SGL cross section curves.

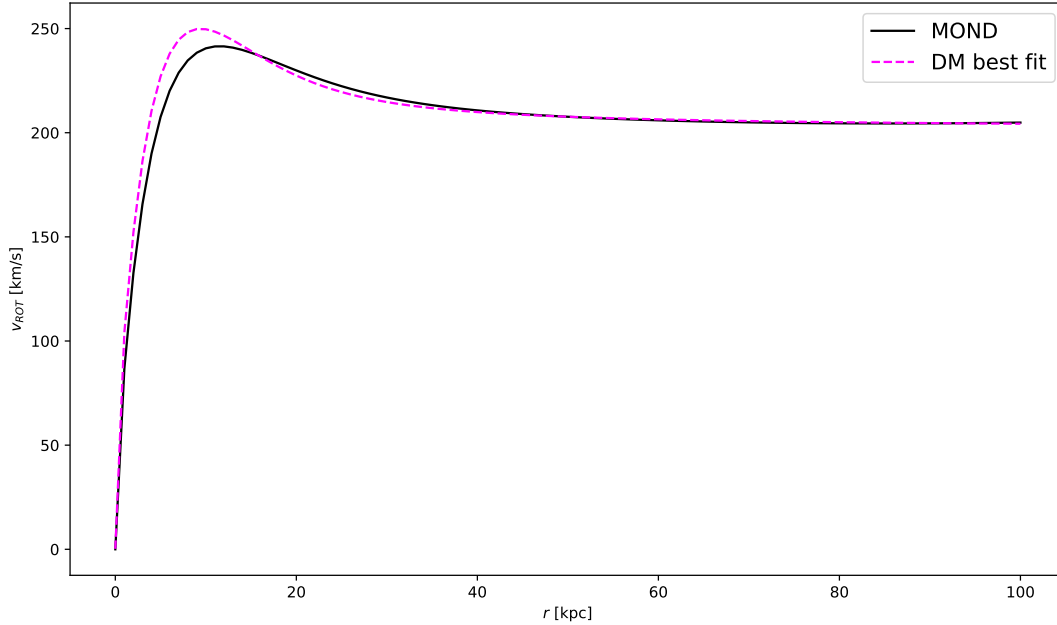


Figure 9 A comparison of MONDian rotation curve predictions with the best-fit DM alternative. The solid black line shows the MONDian rotation curve prediction for a disk galaxy with parameters: $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, and $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The dashed red line shows the corresponding Newtonian fit of a DM halo, which is taken as an NSIS profile with parameters $\rho_0 = 2.51 \cdot 10^7 M_\odot \text{ kpc}^{-3}$, and $r_0 = 5.48$ kpc.

$M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The DM halo parameters are obtained by fitting the Newtonian rotation curve of the DM-baryon system to the MOND-derived rotation curve. Namely, we are “normalizing” the DM model to obtain a rotation curve matching the MOND prediction. Here, to extrapolate the MONDian rotation curve on the galactic plane, v_{ROT} , we numerically derive the MONDian predictions starting from the baryonic and PDM densities of the system (see Appendix B).

Figure 9 shows the MONDian rotation curve and the best fit for the chosen DM halo model, respectively. Figure 8 shows the clear difference between the predictions of the two frameworks, with MOND SGL cross sections being distinctly higher than the DM alternative. While in the DM paradigm inclination effects are present, these are increased in the MOND scenario, as we could expect from the presence of a “phantom dark disk” in the PDM distribution (see Figure 2).

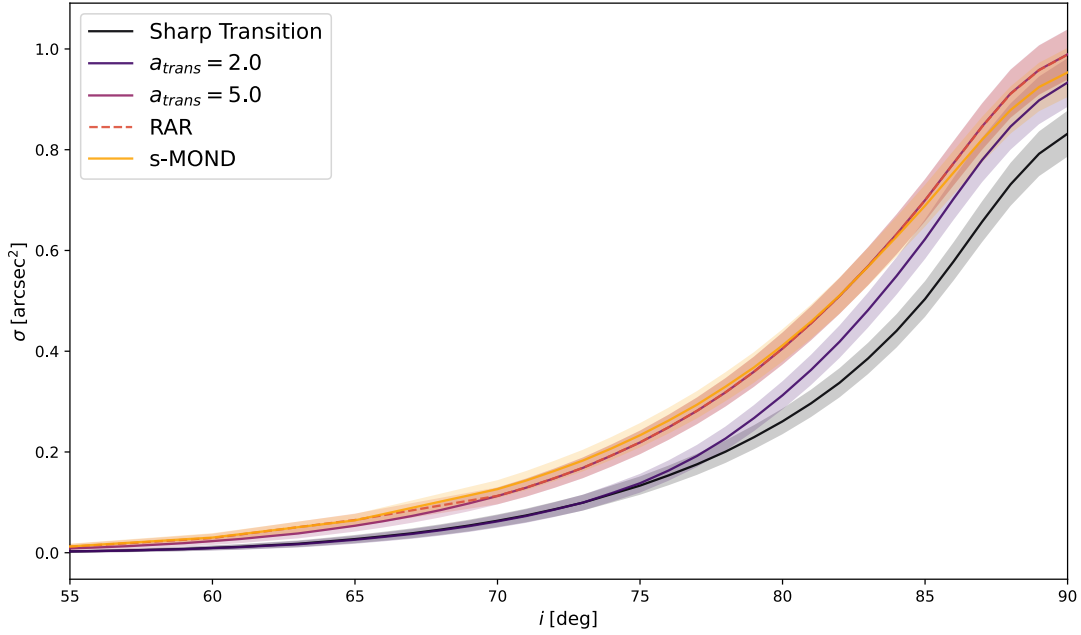


Figure 10 A comparison of lensing cross section predictions with varying inclination angle for the same disk galaxy in MOND for different choices of interpolating functions. For the sigmoid interpolating functions (see Equation 23)), we have chosen $N_{\text{trans}} = 2-5$. The disk galaxy parameters are $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, and $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The shaded areas represent the one standard deviation uncertainties around the obtained SGL cross section curves.

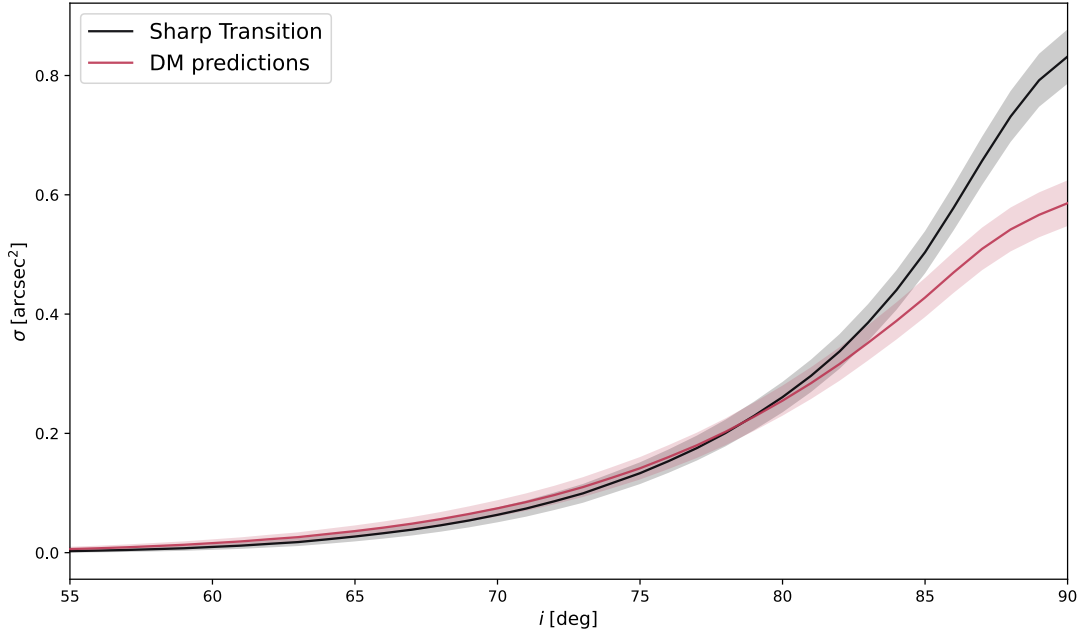


Figure 11 A comparison of lensing cross sections predictions with varying inclination angle, for the same disk galaxy, in the MONDian and the conventional DM scenario. Here, the MOND interpolating function is taken to give an infinitely sharp transition to the dMOND regime. The disk galaxy parameters are $M_d = 10^{11} M_\odot$, $R_d = 3.5$ kpc, $z_d = 0.35$ kpc, and $M_b = 10^9 M_\odot$, and $r_b = 0.7$ kpc. The DM halo is taken as an NSIS profile with parameters $\rho_0 = 2.51 \cdot 10^7 M_\odot \text{ kpc}^{-3}$, and $r_0 = 5.48$ kpc. The lens and source are placed at redshift $z_L = 0.5$ and $z_S = 2.0$, respectively. The shaded areas represent the one standard deviation uncertainties around the obtained SGL cross section curves.

Here we note that, a priori, a different choice of halo model, e.g., either the Navarro–Frenk–White profile (J. F. Navarro et al. 1996a, 1996b), or a Burkert halo (M. Persic & P. Salucci 1990; M. Persic et al. 1996; P. Salucci & A. Burkert 2000; P. Salucci 2019) could produce a different SGL effect. However, we find that changing the DM halo profile does not affect the lensing cross section in a relevant manner if we

maintain a “normalization” of the DM halo with respect to the expected MONDian rotation curve.

Furthermore, as previously discussed, a change in the interpolating function could affect the MONDian results. Therefore, we employ the interpolating functions presented in Equations 22)–24) to test the robustness of the MONDian predictions. Here, Figure 10 compares the MONDian SGL

cross sections obtained with the different interpolating functions. We find that the RAR interpolating and our chosen standard yield perfectly compatible results. Likewise, the sigmoid function, as expected, reproduces the RAR results for high values of the acceleration threshold. Moreover, as the acceleration threshold is lowered, we observe a diminishing SGL effect. This trend is further illustrated by the fact that the infinitely sharp transition interpolating function yields the lowest cross section predictions within the set. This behavior can be understood in terms of the loss of PDM as the transition threshold is lowered. Remarkably, the SGL cross sections predicted by the chosen standard or RAR) interpolating function and those obtained by considering only the dMOND regime differ, at the peak, by about 40% relative to the former. This highlights the importance of the transition regime in determining the SGL produced by disk galaxies.

Finally, as shown in Figure 11, when comparing the results for the infinitely sharp transition interpolating function with those of the DM model, we still find a noticeable difference, further highlighting the relevance of the dMOND regime for SGL with nonspherically symmetric lenses. In particular, we find that deviations from the DM model only become pronounced at higher inclinations with respect to the RAR or our chosen standard MOND predictions, highlighting the influence of the PDM disk on the SGL effect by disk galaxies. Overall, we can thus conclude that disk galaxy SGL can distinguish between the DM and MOND paradigms, regardless of the specific interpolating function of MOND adopted.

5 Conclusions

The DM framework has been highly successful in describing a plethora of phenomena, ranging from astrophysical to cosmological observations. Nonetheless, the nature of DM still eludes the community and currently represents one of the greatest mysteries in physics. As such, alternatives to the DM paradigm, which are able to match a wide variety of observations, warrant investigation. Within the landscape of alternatives to DM, MOND is often considered empirically as the strongest contender to the DM scenario. Indeed, MOND has passed numerous observational tests and also produced a variety of now confirmed predictions.

SGL by galaxies is capable of investigation in both regions of high and low Newtonian accelerations. Therefore, in principle, it represents a valuable probe for MONDian effects over the full range of the acceleration scales within galaxies: from the Newtonian to the deep MOND regime. In particular, studies of disk galaxies would enable a coupling of rotation curve dynamics and SGL. Hence, these represent a great laboratory for investigating gravitational physics.

In this paper, we have derived the SGL effects of disk galaxies in MOND. In the QUMOND formulation of MOND, we have employed the concept of equivalent Newtonian systems to apply the standard lensing formalism to MOND. Within this framework, we obtained the highly nontrivial distribution of phantom DM for disk galaxies and probed its characteristics.

We have found that disk galaxy lensing in MOND produces physically plausible values of deflection angle, magnification, and shear. Furthermore, we have shown that the inclination of the disk galaxy lens with respect to the observer highly influences the lensing geometry. We have seen that within the

MOND paradigm, the classical caustic curves' morphology and number of lensed images expected for disk lenses are readily recovered. Moreover, we have probed the impact of lens morphological and mass parameters on the strong lensing cross section and found nontrivial, and even counterintuitive, correlations due to the onset of purely MONDian effects.

Furthermore, we have drawn comparisons between inclination effects on the lensing cross section predicted in MOND and the DM framework. We have found that MOND nonlinearities dominate over standard Newtonian lensing effects, resulting in: i) steeper inclination effects correlated to the nonspherical phantom DM density profile; and ii) overall larger lensing cross sections. In particular, it is noteworthy that a difference in the cross section of the obtained magnitude (which directly correlates to lensing likelihood) will still clearly impact the number of disk galaxy lenses predicted to be observed in upcoming optical surveys such as Euclid and LSST. Therefore, these future surveys will be able to distinguish the MOND and DM paradigms purely by disk galaxy gravitational lensing number counts.

Remarkably, we find that these results are fairly robust with respect to the choice of interpolating function. In this work, we employed a variety of such functions (see Equations 21)–(24)) to test this robustness, showing that the mismatch between DM and MOND predictions for disk galaxy lenses persists even under extreme choices (e.g., an infinitely sharp transition). Indeed, we find that MOND systematically predicts an enhanced lensing signal for disk galaxies relative to the DM paradigm. However, the magnitude of this enhancement depends on the choice of interpolating function: while the standard and RAR functions yield similar signatures, a sharply varying sigmoid function leads to a reduced lensing prediction. Ultimately, our results emphasize that—for nonspherically symmetric lenses—both the transition regime and the deep MOND limit play a crucial role in the lensing. Indeed, for disk galaxies, we find that the phantom DM disk is ultimately responsible for the boosted lensing effect. Therefore, SGL by disk galaxies provides a powerful astrophysical tool for investigating and constraining the full applicability of MOND.

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Appendix A Phantom Dark Matter Derivation

The full analytic expression for the PDM generated by the combined baryonic density profile given by Equations 14)–17) can be expressed using an equivalent formulation of Equation 7), namely

$$\rho_{\text{ph}} = \frac{1}{4\pi G} \frac{d\tilde{v}}{dy} \nabla y \cdot \nabla \Phi_N \quad (\tilde{v} = y)\rho \quad \text{A1)}$$

where $y = \nabla \Phi_N$, and

$$\|\nabla \Phi_N\|^2 = \|\nabla \Phi_{N,d}\|^2 + \|\nabla \Phi_{N,b}\|^2 = 2 \nabla \Phi_{N,d} \cdot \nabla \Phi_{N,b} \quad \text{A2)}$$

in which the disk and bulge Newtonian potential are given by see also Equations 19) and 20))

$$\Phi_{N,d}(R, z) = -M_d \int_0^\infty dk \frac{J_0(Rk) e^{-|z|k} - z_d k e^{-|z|z_d}}{1 - R_d^2 k^2)^{3/2} (1 - z_d^2 k^2)} \quad \text{A3)}$$

$$\Phi_{N,b}(r) = \frac{M_b}{r_b} \left(1 - \frac{r_b}{\sqrt{r_b^2 + r^2}} \right) \quad \text{A4)}$$

while their gradients, in cylindrical coordinates, are

$$\nabla \Phi_{N,d} = M_d \left(\left[\int_0^\infty dk \frac{J_1(Rk)k}{1 - R_d^2 k^2)^{3/2}} Z(z, k) \right] R \right. \\ \left. z \left[- \int_0^\infty dk \frac{J_0(Rk)}{1 - R_d^2 k^2)^{3/2}} \frac{dZ}{dz} \right] \right) \quad \text{A5)}$$

$$\nabla \Phi_{N,b} = \frac{M_b}{r_b^2} (R \hat{R} - z \hat{z}). \quad \text{A6)}$$

We therefore can directly calculate the norm of the total Newtonian potential. Then, it is possible to directly extract the components of the gradient of the squared norm. We obtain the radial component as

$$\partial_r \|\nabla \Phi_N\|^2 = 2M_b^2 \frac{R_b^2 - 2R^2}{R_b^2 R^2)^4} \\ \times \left(\frac{2M_d^2}{R_b^2} \frac{I_r \partial_r I_r - I_z \partial_r I_z}{R^2)^{3/2}} - \frac{2M_b M_d}{R_b^2} \frac{z^2 - 2r^2}{R^2)^{5/2}} \right) \quad \text{A7)}$$

while the axial component reduces to

$$\partial_z \|\nabla \Phi_N\|^2 = 2M_b^2 \frac{R_b^2 - 2R^2}{R_b^2 R^2)^4} \\ \times \left(\frac{2M_d^2}{R_b^2} \frac{I_r \partial_z I_r - I_z \partial_z I_z}{R^2)^{3/2}} - \frac{2M_b M_d}{R_b^2} \frac{r^2 - 2z^2}{R^2)^{5/2}} \right) \quad \text{A8)}$$

where I_r , I_z , and Z are defined as

$$r = \int_0^\infty dk \frac{J_1(Rk)k}{1 - R_d^2 k^2)^{3/2}} Z(z, k) \\ z = \int_0^\infty dk \frac{J_0(Rk)}{1 - R_d^2 k^2)^{3/2}} \frac{dZ}{dz} \\ Z(z, k) = \frac{e^{-|z|k} - z_d k e^{-|z|z_d}}{1 - z_d^2 k^2}. \quad \text{A9)}$$

One can then use Equations A1)–A9) to calculate the PDM distribution for a given set of disk galaxy parameters and choice of interpolating function.

Appendix B Code Routine

The code routine developed to obtain the results in this work is available at the GitHub repository: https://github.com/chrisarhaw/MOND_lensing. Here, we give a brief explanation of the logic implemented in the three main codes of the routine, i.e., `density_grid.py`, `alpha_rad_tang.py`, `caustic_area.py`, and `rotation_curve.py`.

B 1 `density_grid.py`

This code generates the baryonic and PDM distribution on a (R, z) grid associated with a choice of disk galaxy parameters, and of grid boundaries and steps. To calculate the PDM, we implement the following steps

1. Define the Newtonian potentials for the exponential disk and the spherical Plummer bulge according to Equations A3) and A4).
2. Define the gradients in cylindrical coordinates of the potentials following Equations A5) and A6).
3. Compute the gradient and the norm of the total Newtonian potential by the direct sum of $\nabla \Phi_{N,d}$, and $\nabla \Phi_{N,b}$, i.e., $\nabla \Phi_N = \nabla \Phi_{N,d} + \nabla \Phi_{N,b}$.
4. Compute the gradient of the norm of the total Newtonian by normalizing with respect to $2 \nabla \Phi_N$ Equations A7) and A8).
5. Employ Equation A1) to generate on the grid the PDM distribution once a functional form of the interpolating function is chosen.

Finally, as output, the code also returns the sum of the baryon and PDM densities, multiplied by a weight factor R —accounting for part of the volume element contribution to the calculation of the deflection angle in `alpha_rad_tang.py`, and thus compensating for divergences in the PDM close to the symmetry axis. All of the integration of nonstandard integrals is carried over using the `quad` and `quad_vec` functions of the Python package `integrate` (P. Virtanen et al. 2020). Figure 12 shows the comparison between the PDM density obtained with our code for a Plummer profile ($M_\odot$, and $r_b = 0.7$ kpc), and the respective analytical prediction, for the choice of interpolating function of Equation 21). We find the two to be in good agreement.

B 2 `alpha_rad_tang.py`

This code loads the output of `density_grid.py` and computes the deflection angles' components. To do so, we exploit an interesting formulation of the deflection angle. From

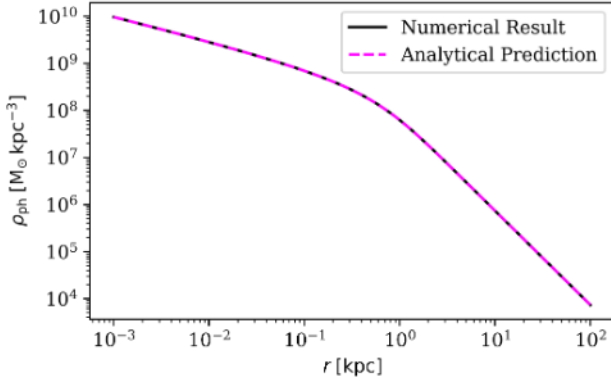


Figure 12 A comparison between the numerically obtained PDM (black line) and analytical prediction (dashed purple line) for a Plummer profile with $M_\odot$ and $r_b = 0.7$ kpc. The MOND interpolating function selected is the one displayed in Equation (21).

Equations (9)–(11) we can write

$$\alpha_{\xi_1 \xi_2} = \frac{2}{c^2} \int_{-\infty}^{\infty} d\xi_3 \iiint \frac{\rho(\xi)}{|\xi - \xi'|} d^3\xi. \quad (B1)$$

Noting that the dependence on ξ_3 within the integral is present exclusively in the geometric term of the integrand (namely, the denominator), we swap the integration order to obtain

$$\alpha_{\xi_1 \xi_2} = \frac{2}{c^2} \iiint \rho(\xi) d^3\xi \int_{-\infty}^{\infty} \frac{d\xi_3}{|\xi - \xi'|}. \quad (B2)$$

The second integral in Equation (B2) is readily computed to obtain the two components of the deflection angle

$$\alpha_{\xi_1} = \frac{2}{c^2} \times \iiint \left(\frac{2 z \sin \lambda - x \cos \lambda - \xi_1}{\xi_1^2 - y - \xi_2^2 - z \sin \lambda - x \cos \lambda)^2 - 2 \xi_1 z \sin \lambda - x \cos \lambda} \right) \rho(\xi) d^3\xi \quad (B3)$$

$$\alpha_{\xi_2} = \frac{2}{c^2} \times \iiint \left(\frac{y - \xi_2}{\xi_1^2 - y - \xi_2^2 - z \sin \lambda - x \cos \lambda)^2 - 2 \xi_1 z \sin \lambda - x \cos \lambda} \right) \rho(\xi) d^3\xi \quad (B4)$$

where λ defines the angle between ξ_3 and the z -axis of the coordinate system adapted to the disk galaxy mass distribution. Then, in this code, we employ Equations (B3) and (B4) to define the components of the deflection angle. We compute these integrals by direct discretization on a grid, whose precision can be specified. From the deflection angle components, the code also computes the inverse magnification matrix and extracts its radial and tangential eigenvalues.

B3 caustic_area.py

This code loads the outputs of `alpha_rad_tang.py`, i.e., the deflection angle components, and the tangential and radial eigenvalues of the inverse magnification matrix, λ' . The radial and tangential critical curves can then be computed by determining the corresponding contours where $\lambda' = 0$, which are traced back into caustic lines using the lens equation (Equation (12)). Once the tangential caustic contours have been found (if they exist), the area within them is calculated. We do this for each density and inclination being investigated, then finally output these values as a numpy array for plotting.

B4 rotation_curve.py

Using a precomputed density field from the `density_grid.py` routine, this code numerically computes the corresponding circular rotation velocity curve on the equatorial plane and then fits an NSIS DM halo profile for subsequent lensing comparisons. To generate the rotation curve, we apply

$$\frac{d\Phi_M}{dr} \Big|_{z=0, \phi=0} = \frac{d}{dr} \int \frac{\rho_M(\vec{r}') d^3\vec{r}'}{|\vec{r}' - \vec{r}|} \quad (B5)$$

where $\rho_M = \rho_B + \rho_{ph}$, and

$$\frac{v_{\text{rot}}^2}{r} = \frac{\Phi_M}{r} \quad (B6)$$

to arrive at the expression for the circular rotation velocity in cylindrical coordinates as

$$v_{\text{rot}}^2 = r \iiint \frac{r' r - r' \cos \phi'}{r'^2 \frac{r'^2}{z'^2 - 2 r r' \cos \phi'}} dr' dz' d\phi' \quad (B7)$$

Once the rotation curve has been calculated for a given disk galaxy, we then fit a curve corresponding to the baryons plus additional nonsingular isothermal DM halo (see Equation (25)).

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References

- Acevedo Barroso, J. A., Riordan, C. M., Clément, B., et al. 2025, *A&A*, **697**, A14
- Allen, R. J., Kacprzak, G. G., Glazebrook, K., et al. 2017, *ApJL*, **834**, L11
- Astesiano, D., Cacciatori, S. L., Gorini, V., & Re, F. 2022, *EPJC*, **82**, 554
- Banik, I., & Zhao, H. 2022, *Symm*, **14**, 1331
- Bartelmann, M. 2010, *CQGra*, **27**, 233001
- Bekenstein, J., & Milgrom, M. 1984, *ApJ*, **286**, 7
- Bekenstein, J. D. 2004, *PhRvD*, **70**, 083509
- Benetti, F., Lapi, A., Gandolfi, G., Salucci, P., & Danese, L. 2023, *ApJ*, **949**, 65
- Beordo, W., Crosta, M., Lattanzi, M., Re Fiorentin, P., & Spagna, A. 2024, *MNRAS*, **529**, 4681
- Bertone, G., & Hooper, D. 2018, *RvMP*, **90**, 045002
- Binney, J., & Tremaine, S. 2008, *Galactic Dynamics: Second Edition* (Princeton, NJ: Princeton Univ. Press)
- Blanco, C., & Leane, R. K. 2024, *PhRvL*, **132**, 261002
- Bosma, A. 1978, PhD thesis, Groningen Univ.
- Bilek, M., Zhao, H., Famaey, B., et al. 2021, *A&A*, **653**, A170
- Candlish, G. N., Smith, R., & Fellhauer, M. 2014, *MNRAS*, **446**, 1060
- Chae, K.-H. 2024, *ApJ*, **960**, 114
- Ciotti, L., Londrillo, P., & Nipoti, C. 2006, *ApJ*, **640**, 741
- Ciotti, L., Zhao, H., & de Zeeuw, P. T. 2012, *MNRAS*, **422**, 2058
- Collett, T. E. 2015, *ApJ*, **811**, 20
- Desmond, H., Hees, A., & Famaey, B. 2024, *MNRAS*, **530**, 1781
- Di Cintio, P., Re, F., & Chiari, C. 2024, *A&A*, **689**, A150
- Dutton, A. A., Brewer, B. J., Marshall, P. J., et al. 2011, *MNRAS*, **417**, 1621
- Famaey, B., & McGaugh, S. S. 2012, *LRR*, **15**, 10
- Forbes, J. C., & Loeb, A. 2018, *MNRAS*, **479**, 171
- Galoppo, M., Re, F., & Wiltshire, D. L. 2025, *CQGra*, **42**, 135004
- Galoppo, M., & Wiltshire, D. L. 2024, arXiv:2406.14157
- Giusti, A. 2020, *PhRvD*, **101**, 124029
- Hernandez, X., Verteletskyi, V., Nasser, L., & Aguayo-Ortiz, A. 2023, *MNRAS*, **528**, 4720
- Hinshaw, G., Weiland, J. L., Hill, R. S., et al. 2009, *ApJS*, **180**, 225
- Keeton, C. R., & Kochanek, C. S. 1998, *ApJ*, **495**, 157
- Ko, C.-M. 2016, *ApJ*, **821**, 111
- Kormann, R., Schneider, P., & Bartelmann, M. 1994, *A&A*, **284**, 285

- Kroupa, P., Gjergo, E., Asencio, E., et al. 2023, in Corfu Summer Inst. 2022, School and Workshops on Elementary Particle Physics and Gravity (Trieste: SISSA)
- Kuzmin, G. G. 1956, *AZh*, 33, 27
- Lang, P., Wuyts, S., Somerville, R. S., et al. 2014, *ApJ*, 788, 11
- Lelli, F., McGaugh, S. S., Schombert, J. M., & Pawlowski, M. S. 2017, *ApJ*, 836, 152
- Liu, J., Chen, X., & Ji, X. 2017, *NatPh*, 13, 212
- Lüghausen, F., Famaey, B., & Kroupa, P. 2015, *CaJPh*, 93, 232
- Magueijo, J., & Smolin, L. 2004, *CQGra*, 21, 1725
- McGaugh, S., Lelli, F., & Schombert, J. 2016, *PhRvL*, 117, 201101
- McGaugh, S. S., Schombert, J. M., Lelli, F., & Franck, J. 2024, *ApJ*, 976, 13
- Meneghetti, M. 2022, *Introduction to Gravitational Lensing: With Python Examples* 1st ed.; Cham: Springer)
- Meneghetti, M., Ragagnin, A., Borgani, S., et al. 2022, *A&A*, 668, A188
- Meng, Y., Wang, Z., Tao, Y., et al. 2021, *PhRvL*, 127, 261802
- Mestel, L. 1963, *MNRAS*, 126, 553
- Milgrom, M. 1983, *ApJ*, 270, 365
- Milgrom, M. 1986, *ApJ*, 306, 9
- Milgrom, M. 2010, *MNRAS*, 403, 886
- Milgrom, M. 2013, *PhRvL*, 111, 041105
- Milgrom, M. 2020, arXiv:2001.09729
- Mistele, T., McGaugh, S., Lelli, F., Schombert, J., & Li, P. 2024a, *JCAP*, 2024, 020
- Mistele, T., McGaugh, S., Lelli, F., Schombert, J., & Li, P. 2024b, *ApJL*, 969, L3
- Moffat, J. W. 2006, *JCAP*, 2006, 004
- Moffat, J. W., & Rahvar, S. 2013, *MNRAS*, 436, 1439
- Mortlock, D. J., & Turner, E. L. 2001, *MNRAS*, 327, 557
- Möller, O., & Blain, A. W. 1998, *MNRAS*, 299, 845
- Navarro, J. F., Eke, V. R., & Frenk, C. S. 1996a, *MNRAS*, 283, L72
- Navarro, J. F., Frenk, C. S., & White, S. D. M. 1996b, *ApJ*, 462, 563
- Oppenheim, J., Panella, E., & Pontzen, A. 2024, arXiv:2407.13820
- Oppenheim, J., & Weller-Davies, Z. 2023, arXiv:2302.07283
- Orban de Xivry, G., & Marshall, P. 2009, *MNRAS*, 399, 2
- Persic, M., & Salucci, P. 1990, *MNRAS*, 245, 577
- Persic, M., Salucci, P., & Stel, F. 1996, *MNRAS*, 281, 27
- Planck Collaboration, Aghanim, N., Akrami, Y., et al. 2020, *A&A*, 641, A6
- Pouliasis, E., Matteo, P. D., & Haywood, M. 2017, *A&A*, 598, A66
- Profumo, S., Giani, L., & Piattella, O. F. 2019, *Univ*, 5, 213
- Re, F., & Di Cintio, P. 2023, *A&A*, 678, A110
- Rubin, V. C., Ford, W. K. J., & Thonnard, N. 1978, *ApJL*, 225, L107
- Salucci, P. 2019, *A&ARv*, 27, 60
- Salucci, P., & Burkert, A. 2000, *ApJL*, 537, L9
- Sanders, R. H. 2014, *MNRAS*, 439, 1781
- Sanders, R. H., & Land, D. D. 2008, *MNRAS*, 389, 701
- Scarpa, R., Falomo, R., & Treves, A. 2022, *MNRAS*, 512, 544
- Schneider, P., Kochanek, C. S., & Wambsganss, J. 2006, *Gravitational Lensing: Strong, Weak and Micro* 1st ed.; Berlin: Springer)
- Shan, H. Y., Feix, M., Famaey, B., & Zhao, H. 2008, *MNRAS*, 387, 1303
- Sofue, Y., & Rubin, V. 2001, *ARA&A*, 39, 137
- Sotiriou, T. P., & Faraoni, V. 2010, *RvMP*, 82, 451
- Stiskalek, R., & Desmond, H. 2023, *MNRAS*, 525, 6130
- Suyu, S. H., Hensel, S. W., McKean, J. P., et al. 2012, *ApJ*, 750, 10
- Tihhonova, O., Courbin, F., Harvey, D., et al. 2018, *MNRAS*, 477, 5657
- Treu, T. 2010, *ARA&A*, 48, 87
- Trott, C. M., Treu, T., Koopmans, L. V. E., & Webster, R. L. 2010, *MNRAS*, 401, 1540
- van der Kruit, P. C., & Searle, L. 1981, *A&A*, 95, 105
- van der Kruit, P. C., & Searle, L. 1982, *A&A*, 110, 61
- Verlinde, E. 2011, *JHEP*, 2011, 29
- Verlinde, E. P. 2017, *ScPP*, 2, 016
- Virtanen, P., et al. 2020, *NatMe*, 17, 261
- Wang, Y., & Turner, E. L. 1997, *MNRAS*, 292, 863
- Wells, P., Fassnacht, C. D., & Rusu, C. E. 2023, *A&A*, 676, A95
- Wu, X., Famaey, B., Gentile, G., Perets, H., & Zhao, H. 2008, *MNRAS*, 386, 2199
- Zhu, Y., Ma, H.-X., Dong, X.-B., et al. 2023, *MNRAS*, 519, 4479